



ESG-washing detection in corporate sustainability reports

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ABSTRACT

This study introduces the ESG-washing Severity Index (ESGSI) for quantitatively assessing discrepancies between portrayed and actual sustainability practices in corporate disclosures. Using advanced Natural Language Processing (NLP) techniques, we analyze sustainability reports from 749 listed companies, integrating sentiment analysis with the frequency of sustainability terms to calculate the ESGSI. Our findings reveal significant variation in ESG-washing practices across industries and geographical regions. The ESGSI serves as a critical tool for stakeholders and policymakers, highlighting the need for stricter sustainability reporting standards and more effective regulatory frameworks to combat ESG-washing. This study contributes to the growing body of literature on corporate sustainability and provides practical implications for investors, corporate managers, and policy-makers in evaluating and improving ESG practices and disclosures.

1. Introduction

In the context of responsible business practices, the assessment of a company's Environmental, Social, and Governance (ESG) performance and disclosure is paramount for stakeholders. ESG performance evaluates the tangible actions, policies, and outcomes related to a company's ESG practices, encompassing factors such as environmental impact, social engagement, and governance structures (Smith & Johnson, 2020).

Conversely, the ESG disclosure score measures the transparency with which a company communicates its ESG information to stakeholders. This score highlights both the quality and quantity of information present in ESG reports and other communication channels (Jones & Brown, 2019). Both metrics are essential in evaluating a company's sustainability efforts.

The rise of sustainability as a focal point in consumer choices and investment decisions has amplified the importance of accurately measuring and combating greenwashing practices (C. Liu et al., 2024; Teti et al., 2024; Z. Liu & Li, 2024).

As reported by de Freitas Netto et al. (2020), the term greenwashing was first coined by environmentalist Jay Westerveld in 1986¹ and it refers to market practices, both intentional and unintentional, whereby the publicly disclosed sustainability profile of an entity or financial product and its actual sustainability profile diverge. This can occur at entity level, product level, or service level across the sustainable value chain (ESMA, 2024). The phenomenon ranges from outright

manipulation of disclosures (the most extreme case) to more subtle misleading practices that overstate a company's environmental efforts (Delmas & Burbano, 2011; Lyon & Montgomery, 2015).

The implications of greenwashing for investors and stakeholders are significant and multifaceted. For investors, greenwashing can lead to misallocation of capital, as they may unknowingly support companies that do not align with their sustainability goals (Kim & Lyon, 2015). This misalignment can result in reputational risks and potential financial losses when the truth is revealed. For stakeholders, including consumers and employees, greenwashing erodes trust in corporate sustainability initiatives, potentially leading to skepticism towards genuine efforts and hampering overall progress in sustainable development (Delmas & Burbano, 2011). Moreover, greenwashing can distort market competition, disadvantaging companies that are truly committed to sustainability (Lyon & Maxwell, 2011).

In this paper we introduce a measure for detecting the ESGwashing Severity Index (ESGSI), related not only to the Environmental dimension of ESG practices, but also to the Social and Governance dimension. The dichotomy between ESG performance and ESG disclosure lies at the heart of the ESGwashing issue. While ESG performance evaluates a company's tangible actions and outcomes in environmental conservation, social responsibility, and governance practices, ESG disclosure assesses the transparency and accuracy of how a company communicates its ESG information to stakeholders (Miles & Covin, 2021). Dhalliwai et al. (2012) emphasized that firms with higher ESG performance

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¹ Westerveld, 1986

are more likely to engage in ESG disclosure. However, there's a debate on whether better ESG disclosure leads to improved ESG performance or vice versa (Cheng et al., 2019; Kim et al., 2018).

The regulatory landscape for ESG disclosure practices varies significantly across countries, with a spectrum ranging from voluntary to mandatory regulations. For instance, the European Union has implemented the Sustainable Finance Disclosure Regulation (SFDR), which mandates specific ESG disclosures for financial market participants (European Parliament and Council of the European Union, 2019). Specifically, it aims to combat ESGwashing by requiring detailed ESG disclosures from financial market participants, making it harder for companies to misrepresent their sustainability practices. In contrast, with a focus on the Environment, the United States has historically relied more on voluntary disclosures, although the Securities and Exchange Commission (SEC) has recently proposed rules to enhance and standardize climate disclosures (U.S. Securities and Exchange Commission, 2022). The SEC's proposed climate disclosure rules in the US represent a shift towards more mandatory reporting, potentially reducing opportunities for greenwashing by standardizing how companies report their environmental impact and initiatives. These varying approaches significantly impact the prevalence and nature of ESGwashing, highlighting the need for a more harmonized global regulatory framework (Hong et al., 2020).

We use an international sample of listed firms operating in different industries. Employing artificial intelligence and Natural Language Processing (NLP) techniques, our methodology aims to analyze textual data from ESG disclosure documents, identifying linguistic and contextual indicators of ESGwashing. Through text classification and machine learning models, it will provide a comprehensive assessment of ESGwashing severity (e.g. ESGSI), thereby enabling stakeholders to differentiate between genuine corporate sustainability efforts and deceptive practices (Kleffel & Muck, 2023; Liu & Bohnsack, 2022; Perez-Baltres, 2021).

The outcomes of our investigation expose substantial disparities between ESG performance and disclosure in numerous instances, signaling the prevalence of ESGwashing practices. This disconnect between declared commitments and practical implementations underscores the pressing need to address ESGwashing (Chatterji & Toffel, 2019; Wang & Marquis, 2020).

The ESGSI has been put to use in gauging the extent of ESGwashing in sustainability reporting. The outcomes indicate that a noteworthy proportion of companies exhibit notable levels of ESGwashing, accentuating the requirement for heightened scrutiny and regulatory measures in this domain (Delmas & Burbano, 2011; Khan & Serafeim, 2021).

Furthermore, the research sheds light on the intricate nature of ESGwashing, which is multifaceted within the realm of corporate sustainability. The identification and mitigation of ESGwashing call for unwavering vigilance, comprehensive assessments, and ongoing scrutiny. Regulatory frameworks provide essential guidance in mitigating ESGwashing, fostering truthful reporting, and shaping responsible corporate conduct (Vanclay et al., 2020).

This paper contributes to the existing literature in several ways. First, it provides a comprehensive analysis of the relationship between ESG performance and ESG disclosure, offering new insights into the mechanisms of ESGwashing. Second, it introduces the ESGSI, which can serve as a valuable tool for investors, regulators, and other stakeholders in assessing the authenticity of corporate sustainability claims. Finally, it offers policy recommendations for enhancing regulatory frameworks to combat ESGwashing more effectively.

2. Literature review

Recent research has highlighted the complex relationship between ESG performance and disclosure, revealing nuanced impacts on firm value and stakeholder perceptions. Grewal et al. (2020) examined the real effects of mandatory ESG disclosure, finding that firms subject to

such requirements experienced significant improvements in their ESG performance. This suggests that disclosure regulations can drive tangible changes in corporate behavior, beyond mere reporting.

The quality of ESG disclosure has emerged as a critical factor in its effectiveness. Berg et al. (2022) introduced the concept of "aggregate confusion" in ESG ratings, demonstrating substantial divergence among major ESG rating providers. This inconsistency highlights the challenges in measuring and communicating ESG performance effectively, potentially leading to misallocation of capital in ESG-focused investments.

The integration of ESG factors into corporate strategy and risk management has also evolved. Krueger et al. (2020) surveyed institutional investors globally, revealing that climate risks have financial implications for portfolio firms and that these risks are beginning to be priced in financial markets. This finding emphasizes the growing importance of robust ESG disclosures and their impact on investment decisions.

Technological advancements have opened new avenues for ESG analysis. Cao et al. (2021) employed machine learning techniques to develop a novel measure of corporate culture based on employee reviews. This approach offers new perspectives on measuring intangible aspects of ESG performance, such as workplace culture and employee satisfaction, which are often challenging to quantify through traditional disclosure methods.

Significant challenges remain in aligning ESG performance with disclosure practices. Christensen et al. (2021) provide a comprehensive review of mandatory CSR and sustainability reporting regulations, highlighting the need for more standardized approaches to ESG disclosure to enhance comparability and reduce information asymmetry. Lagasio and Cucari (2019) conducted a meta-analytical review of corporate governance and ESG disclosure, emphasizing the need for more sophisticated measurement techniques to capture the multifaceted nature of ESG performance and its communication.

These findings collectively underscore the evolving landscape of ESG performance and disclosure, highlighting the need for more sophisticated measurement tools and standardized reporting frameworks to bridge the gap between corporate sustainability efforts and stakeholder perceptions. As ESG factors have gained prominence in investment decisions, the potential for ESGwashing has increased, leading to a need for more robust methods of assessing corporate sustainability claims (Lyon & Montgomery, 2015).

Most of the extant literature is focused on the Environmental dimension of ESG and thus on the greenwashing issue. The implications of greenwashing extend beyond reputational damage. Kim and Lyon (2015) found that firms strategically disclose positive environmental information while withholding negative information, a practice they term "brownwashing". This selective disclosure can lead to an overestimation of a firm's environmental performance and potentially misguide stakeholder decisions.

Moreover, greenwashing can have significant financial implications. Krueger (2015) found that investors react negatively to positive CSR news when they perceive it as inconsistent with shareholder value maximization. This highlights the importance of aligning ESG initiatives with overall corporate strategy and communicating them authentically.

The challenge of accurately measuring and quantifying greenwashing has been a significant focus of recent research. Traditional methods often relied on manual content analysis of sustainability reports and corporate communications. However, advancements in data analytics and artificial intelligence have opened new avenues for more sophisticated analysis.

Loughran et al. (2009) demonstrated the potential of textual analysis in assessing corporate disclosures, including sustainability reports. Their work highlights the importance of considering the tone and content of corporate communications in evaluating ESG performance and potential greenwashing.

More recently, machine learning techniques have been employed to develop novel measures of corporate sustainability practices. For

instance, [Cao et al. \(2021\)](#) used machine learning to develop a measure of corporate culture based on employee reviews, offering a new perspective on measuring intangible aspects of ESG performance.

Still in the context of climate disclosures, [Engle et al. \(2020\)](#) introduced an innovative approach to measuring climate risk, developing mimicking portfolios that can be used to hedge climate change news. Their work demonstrates the increasing sophistication of ESG financial instruments and the potential for quantitative methods in assessing the authenticity of climate claims.

The prevalence and sophistication of greenwashing practices have not only raised concerns among stakeholders but have also highlighted significant gaps in current regulatory frameworks. As [Delmas and Burbano \(2011\)](#) argue, the lack of standardized definitions and metrics for sustainability claims creates an environment where greenwashing can thrive. This regulatory vacuum has led to increased calls for more robust and comprehensive ESG disclosure requirements.

Regulators face the challenging task of developing standards that can effectively curtail greenwashing while promoting genuine sustainable practices. [Christensen et al. \(2021\)](#) provide a comprehensive review of the mandatory ESG disclosure regulations across the globe, finding that such regulations have led to increased disclosure and improved ESG performance among affected firms. However, the effectiveness of these regulations in curbing greenwashing remains a subject of debate. The challenge for regulators lies in striking a balance between encouraging transparent and meaningful ESG disclosure while avoiding overly burdensome requirements that might discourage firms from engaging in sustainable practices. Moreover, given the global nature of many corporations and financial markets, there is a growing need for harmonized international standards for.

2.1. ESG reporting and anti-greenwashing measures

As the field evolves, there is a clear need for more sophisticated tools and metrics to distinguish between genuine sustainability efforts and ESGwashing. This study aims to contribute to this effort by introducing the ESGSI, a novel metric that combines sentiment analysis with sustainability content assessment to provide a more nuanced evaluation of corporate sustainability communications.

3. Methodology

Our methodology integrates advanced NLP techniques with quantitative scoring mechanisms to assess ESGwashing in corporate sustainability reports. This approach builds upon previous work in textual analysis of financial documents ([Loughran & McDonald, 2011](#)) and the application of machine learning to ESG reporting. We begin by manually collecting and preprocessing standalone ESG reports from 749 globally listed companies, published in 2023. This process, inspired by the data preparation methods of [Mihalcea and Tarau \(2004\)](#), ensures a consistent and comparable dataset.

The core of our analysis relies on three NLP techniques: Term Frequency-Inverse Document Frequency (TF-IDF) vectorization, Latent Dirichlet Allocation (LDA), and sentiment analysis. TFIDF, following the approach of [Engle et al. \(2020\)](#), allows us to identify key sustainability indicators within the reports. LDA, as implemented by [Blei et al. \(2003\)](#), uncovers latent thematic structures, providing insights into the central topics of discussion. Sentiment analysis, using methods similar to those of [Loughran and McDonald \(2016\)](#), assesses the emotional tone of the reports.

The culmination of our methodology is the development of the ESGSI. This novel metric combines sentiment scores with sustainability content quantification, drawing inspiration from the greenwashing measurement approaches of [Delmas and Burbano \(2011\)](#) and [Lyon and Montgomery \(2015\)](#). By normalizing and comparing these scores, we create a robust measure of potential ESGwashing practices.

Our approach aims to provide a comprehensive assessment of

sustainability reports, offering stakeholders deeper insights into a company's ESG commitments while simultaneously detecting potential ESGwashing. This methodology enables a holistic perspective on corporate sustainability practices, contributing to the growing body of literature on ESG disclosure and corporate social responsibility.

3.1. Data

The initial sample included the top 2000 companies in terms of market capitalization listed worldwide, retrieved from Refinitiv. To ensure global coverage, data were collected from companies operating in various geographical regions and multiple sectors, including manufacturing, technology, energy, healthcare, and finance.

We manually collected standalone ESG reports published in 2023, which resulted in a significant reduction of our sample size from 2000 to 749 companies. This reduction was necessary to maintain consistency and comparability in our analysis. Specifically, we applied several criteria for inclusion in our final sample. Only reports published in English were considered to ensure consistent application of our natural language processing techniques. All reports had to pertain to the 2023 reporting year to provide a contemporaneous snapshot of ESG disclosure practices. We included only standalone ESG reports; companies that integrated their ESG disclosure within their annual financial reports were excluded to maintain structural consistency across our sample and to focus solely on dedicated sustainability reporting. Additionally, reports had to be complete and accessible for comprehensive analysis.

This manual collection and filtering process, while reducing our sample size, ensured a high level of consistency and comparability across the analyzed reports. It's worth noting that for the purposes of our study, which aims to define and calculate the ESGSI, a cross-sectional dataset focused on a single year (2023) is sufficient. Our methodology does not require a panel dataset spanning multiple years, as the index calculation is designed to provide a robust measure of potential ESGwashing based on a single, comprehensive ESG report. This approach allows us to capture the most recent ESG disclosure practices and potential ESGwashing tendencies in a rapidly evolving regulatory and business environment.

3.1.1. Data preprocessing

Our study involves a systematic approach to preprocess textual data extracted from ESG reports. The methodology employs Python for automation of these processes, involving string manipulation, file handling, regular expressions, and natural language processing. We use the Natural Language Toolkit (NLTK) library ([Bird et al., 2009](#)), and lemmatization using NLTK's WordNetLemmatizer. We use NLTK version 3.6.2 for tokenization, stopword removal, and lemmatization. Scikitlearn version 0.24.2 is employed for text vectorization and machine learning tasks ([Pedregosa et al., 2011](#)). We augment NLTK's default English stopwords with a custom list, which includes additional terms frequently found in corporate literature. These stopwords, along with specified punctuation marks, are intended to be removed from the text to refine the dataset. The preprocessing function encapsulates several key steps:

1. **URL Removal:** This step eliminates web links from the text, which are irrelevant to the analysis.
2. **Normalization:** All text is converted to lowercase to ensure uniformity. Numeric characters and punctuation are removed using regular expressions and string manipulation techniques.
3. **Tokenization and Cleaning:** The text is broken down into individual words or tokens. Subsequently, tokens that are stopwords or shorter than three characters are removed to eliminate trivial content.
4. **Lemmatization:** Words are converted to their base or lemma form to reduce the data's complexity by consolidating different forms of the same word.

The final sample covers 749 companies listed worldwide.

3.2. Text analysis using natural language processing (NLP)

3.2.1. TF-IDF

The analysis of ESGwashing in sustainability reports hinges on identifying key sustainability indicators categorized into three domains: Environmental, Social, and Governance (ESG). These categories are populated with specific keywords relevant to each sustainability aspect.

We employ Term Frequency-Inverse Document Frequency (TF-IDF) vectorization to convert text data into numerical values, following the approach of Engle et al. (2020). We use Scikit-learn's TfidfVectorizer with settings chosen to balance comprehensiveness and computational efficiency, in line with best practices in NLP research. These settings include ignoring terms that appear in more than 95 % of documents, only considering terms that appear in at least 2 documents, limiting the vocabulary to the top 10,000 terms, and considering both unigrams and bigrams. This method allows for the evaluation of the relative importance of each term within the documents against the entire dataset. The TF-IDF vectorizer, provided by the scikit-learn library, is used with adjustable settings to accommodate the unique features of different sustainability reports.

The process begins by establishing detailed lists of sustainability indicators for governance, environmental impact, and social impact. These lists include terms like 'Diversity', 'Carbon Emissions', and 'Equal Employment Opportunities', aiming to cover a wide range of themes frequently discussed in sustainability reports. These indicators are based on established ESG reporting frameworks such as Global Reporting Initiative (GRI) Standards (Global Reporting Initiative, 2021) and Sustainable Accounting Standards Board (SASB) Standards (Sustainability Accounting Standards Board, 2018).

Documents are preprocessed and stored in a directory, from which they are retrieved and transformed into a list of texts for analysis. Each document is then vectorized with the TF-IDF model to calculate a score for each term based on its frequency relative to other documents. This approach highlights terms that are particularly significant within specific reports, thereby pinpointing the sustainability indicators most prominently featured in the corporate ESG disclosures.

3.2.2. LDA

We apply Latent Dirichlet Allocation (LDA) to explore thematic structures within sustainability reports, following the methodology of Blei et al. (2003). LDA reveals the underlying themes that might not be immediately obvious, providing insights into the central topics of discussion in the reports, including those related to ESGwashing. In our study, we adopt a LDA model to unearth latent thematic structures within the corpus of preprocessed textual documents. The procedure commences by segmenting each document into its constituent words, thus preparing the dataset for the ensuing topic modeling steps. We use the Gensim library (Rehurek & Sojka, 2010) for LDA implementation, with key steps including text preparation, dictionary construction, noise reduction, and LDA model training. We train the LDA model with 10 topics, using 100 training passes and alpha='auto'. The number of topics was chosen based on interpretability and coherence scores, following the approach of Roder et al. (2015).

We incorporate sentiment analysis using the TextBlob library (Loria, 2018) to calculate sentiment scores for each document. This approach has been validated in previous studies on financial text analysis (Loughran & McDonald, 2011). The corpus, a collection of documents represented as strings of preprocessed text, is tokenized into lists of words. Utilizing the gensim library's corpora.Dictionary, a mapping between words and their unique integer identifiers is established. The dictionary is pruned by eliminating extremes. Words appearing in less than 15 documents and those present in more than 50 % of the corpus are filtered out, while the maximum vocabulary size is capped. The refined dictionary is translated into a Bag-of-Words (BoW) corpus, with

documents rendered as vectors of word frequencies. The LDA model parameters are configured, specifying the number of topics, the number of training passes, and other hyperparameters essential for the algorithm. Finally, the LDA model is trained on the BoW corpus. The id2word parameter is set to facilitate the interpretation of the topics discovered by the model.

The following table (Table 1) reports the output of the LDA model application. This table presents a succinct summary of the top five terms within each of the ten topics derived from an LDA topic model. Each column corresponds to a specific topic, labeled from Topic 0 through Topic 9, with the terms and their respective weights (in parentheses) listed in the rows below. An additional row at the bottom offers a brief interpretation of each topic, suggesting the overarching thematic focus based on the dominant terms.

3.2.3. Sentiment analysis

The study enhances its analysis of ESGwashing in sustainability reports by incorporating sentiment analysis using the TextBlob library to calculate sentiment scores for each document. This analysis is crucial for understanding the overall emotional tone of the reports, particularly how positive sentiments are used strategically by companies to influence stakeholders.

By integrating sentiment analysis with sustainability content quantification, the methodology assesses potential ESGwashing in corporate communications. It combines sentiment scores with TF-IDF scores of sustainability terms to objectively evaluate how extensively sustainability topics are discussed in the reports.

Both sentiment and sustainability scores are normalized to a uniform scale to ensure comparability and address potential biases due to variations in document length and linguistic style. The ESGSI is then derived by subtracting the normalized sustainability scores from the normalized sentiment scores. A positive index indicates potential ESGwashing, where positive sentiment exceeds substantive content on sustainability, while a negative or zero index suggests more authentic sustainability communications.

This approach not only measures the discrepancy between corporate claims and actual sustainability reporting but also provides a valuable metric for evaluating the authenticity of these claims.

3.3. ESGwashing severity index (ESGSI)

The ESGSI is calculated by subtracting the normalized sustainability score from the normalized sentiment polarity score. This approach builds on previous work on measuring corporate greenwashing (Delmas & Burbano, 2011; Lyon & Montgomery, 2015). To ensure comparability, both sentiment and sustainability scores are normalized to a uniform scale using min-max normalization. This normalization technique is widely used in text analysis and machine learning to address potential biases due to variations in document length and linguistic style (Sebastiani, 2002).

$$ESGSI = \frac{SEN_i - \overline{SEN}}{\sigma_{SEN}} - \frac{SUS_i - \overline{SUS}}{\sigma_{SUS}}$$

Where:

SEN_i : sentiment score for company i .

$\overline{SEN}$: mean sentiment score across all companies

σ_{SEN} : standard deviation of sentiment scores

SUS_i : sustainability score for company i

$\overline{SUS}$: mean sustainability score across all companies

σ_{SUS} : standard deviation of sustainability scores

This index attempts to capture discrepancies between the sentiment of the communication and the actual emphasis on sustainability. If the sentiment polarity is higher than the sustainability content score, suggesting a positive ESGSI, it indicates that the sentiment might be overly positive compared to the actual sustainability content. This could

Table 1
Summary of LDA Topic Modeling Results.

Topic	Word 1	Word 2	Word 3	Word 4	Word 5	Interpretation
0	energy (0.0104)	emission (0.0091)	sustain. (0.0088)	management (0.0077)	water (0.0076)	Environment
1	risk (0.0274)	climate (0.0156)	investment (0.0155)	sustainability (0.0143)	esg (0.0125)	Environment
2	new (0.0053)	employee (0.0052)	also (0.0050)	business (0.0045)	group (0.0044)	Human Resource
3	financial (0.0121)	share (0.0110)	board (0.0090)	million (0.0069)	statement (0.0067)	Finance
4	employee (0.0139)	business (0.0096)	sustain. (0.0090)	esg (0.0085)	data (0.0078)	Environment
5	sustainability (0.0140)	product (0.0139)	material (0.0114)	progress (0.0090)	number (0.0087)	Environment
6	risk (0.0217)	impact (0.0145)	customer (0.0133)	sustain. (0.0098)	zurich (0.0093)	Environment
7	management (0.0085)	board (0.0072)	performance (0.0064)	risk (0.0062)	financial (0.0061)	Finance
8	employee (0.0093)	program (0.0073)	community (0.0065)	business (0.0065)	product (0.0055)	Human Resource
9	financial (0.0139)	risk (0.0116)	bank (0.0105)	asset (0.0103)	value (0.0088)	Finance

Note: Top five weighted terms for each of the ten topics identified by LDA. The rightmost column provides an automated interpretation of the predominant theme for each topic.

suggest potential ESGwashing, where the communication may be attempting to appear more sustainable or responsible than it is substantively supported by the document’s content. Conversely, a non-positive index suggests that the sentiment aligns with or is less than the factual sustainability content, indicating likely genuine sustainability efforts.

The script labels each document based on its ESGSI, categorizing them as either “Potential ESGwashing” or “Likely Genuine.” These labels help stakeholders, including investors, regulators, and the public, discern the authenticity of a company’s sustainability claims.

To ensure the reliability and validity of our analysis, we implement several measures. For the manual data collection process, we employ two independent raters and calculate Cohen’s Kappa to assess agreement (Cohen, 1960). We use 5-fold cross-validation to evaluate the robustness of our LDA and sentiment analysis models (Kohavi, 1995). We also conduct sensitivity analyses by varying key parameters (e.g., number of LDA topics, TF-IDF settings) to test the robustness of our results. These measures align with best practices in NLP and machine learning research (Goodfellow et al., 2016).

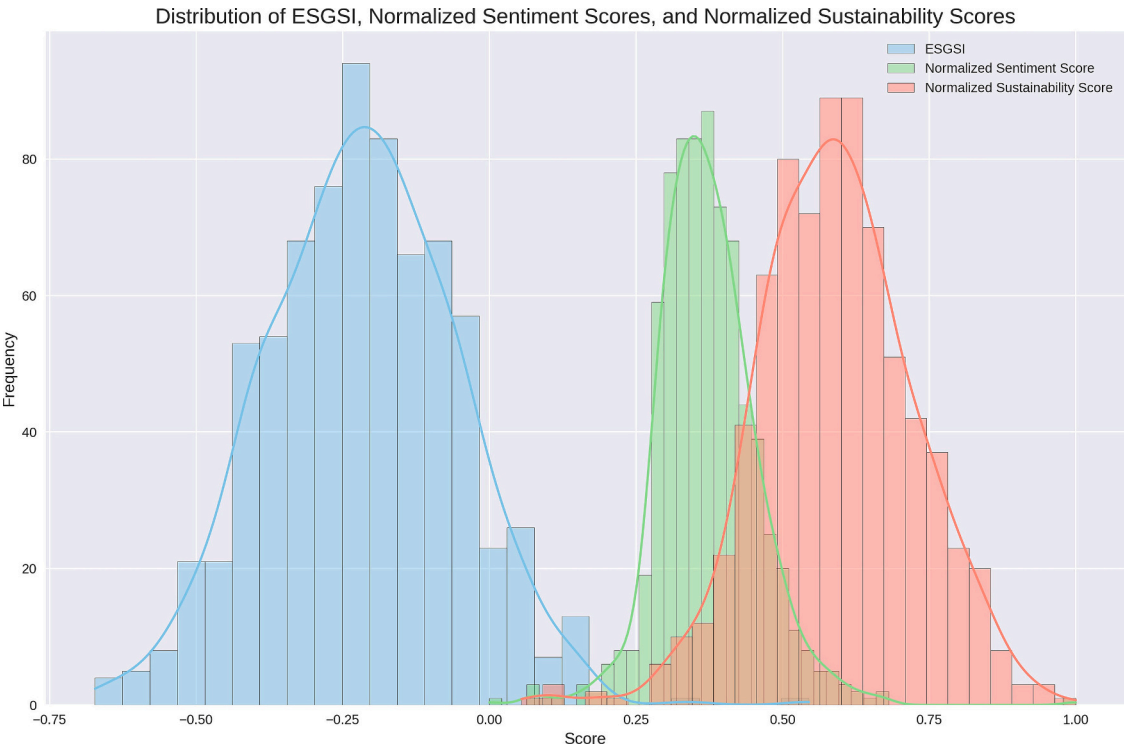


Fig. 1. Frequency distributions.
Note: This figure illustrates the distribution of the ESGSI across the dataset. The ESGSI provides insights into the extent of ESGwashing practices among various entities. Higher values may indicate a higher level of ESGwashing, while lower values suggest a lower degree of ESGwashing. The figure also illustrates the distribution of Normalized Sentiment Scores and Normalized Sustainability Scores. The scores are distributed over a continuous range from 0 to 1.

4. Results

In this section, we present the results of our analysis of sustainability reports. Fig. 1 depicts the distribution of the ESGSI. The histogram portrays the frequency distribution of the ESGSI among the evaluated corporate sustainability documents. The x-axis represents the ESGSI, which is computed as the normalized difference between sentiment polarity and sustainability scores. The y-axis displays the number of documents exhibiting corresponding index values. The distribution is bell-shaped and centered around an ESGSI of zero, reflecting a prevalence of documents whose positive sentiment is proportionate to their sustainability content. Notably, a symmetric dispersion pattern around the central value indicates an equivalent propensity for documents to exhibit both potential greenwashing behaviors and genuine sustainability representation. The presence of fewer documents in the tails of the distribution suggests that extreme cases of sentiment-substance mismatch, whether overly positive or negative, are relatively uncommon in the dataset. This symmetry may point to the nuanced complexities of corporate sustainability communication, wherein a significant proportion of documents maintain a balance between how they portray their sustainability efforts and the actual emphasis on sustainable practices within their content.

The histogram in Fig. 1 also illustrates the distribution of normalized sentiment scores. The scores are distributed over a continuous range from 0 to 1, with the majority of data points clustering around the central values. The overlaying curve suggests an approximation to the normal distribution, indicating the sentiment scores fall into a typical bell-curve pattern. The apparent symmetry of the distribution around the central peak implies that the sentiment is as likely to be positive as it is to be negative. However, the presence of outliers, particularly towards the lower end of the scale, may indicate instances of extremely negative sentiment that could warrant further investigation. Assuming the data is representative of a broader population, one might infer that the sentiments expressed towards the subject are generally moderate, neither overly positive nor negative. This could suggest that public opinion is relatively balanced or that polarized sentiments are evening out in the aggregation. The frequency distribution of the Normalized Sustainability Scores that appears to be skewed towards higher values, indicating a tendency towards positive sustainability reporting.

The descriptive statistics of the dataset are summarized in Table 2, which presents the mean, standard deviation, minimum, and maximum values for each of the variables considered in our study. The ESGSI

Table 2
Descriptive Statistics.

	ESGSI ^a	Normalized Sustainability Score ^b	Normalized Sentiment Score ^c	Sentiment Polarity ^d	Sentiment Subjectivity ^e
count	749	749	749	749	749
mean	-0.2179	0.5906	0.3727	0.0963	0.3506
std	0.1614	0.1321	0.0853	0.0275	0.0408
min	-0.6726	0.0546	0.0000	-0.0237	0.0000
25 %	-0.3245	0.5020	0.3188	0.0790	0.3321
50 %	-0.2177	0.5876	0.3668	0.0944	0.3534
75 %	-0.1054	0.6732	0.4169	0.1105	0.3722
max	0.5449	10.000	10.000	0.2982	0.4935

^a ESGSI: Indicates perceived or reported ESGwashing activities. Negative values suggest low levels of ESGwashing.

^b Normalized Sustainability Score: Scores range from 0 (low sustainability) to 1 (high sustainability), representing the extent of sustainable practices.

^c Normalized Sentiment Score: Measures the overall sentiment towards the entities' sustainability practices, normalized between 0 and 1.

^d Sentiment Polarity: Reflects the tone of sentiment, ranging from negative to positive.

^e Sentiment Subjectivity: Indicates the level of objectivity in sentiment expressions, where 0 is completely objective and higher values are more subjective.

predominantly shows negative values, suggesting a general absence of greenwashing practices among the entities studied. The Normalized Sustainability Score varies across the dataset, with some entities scoring as high as 1, which suggests excellent sustainability practices, and others scoring very low, highlighting significant room for improvement.

The data indicates a general trend of moderate sustainability engagement across the observed entities, with considerable variability in how these practices are perceived. The range in the ESGSI and Normalized Sentiment Score underscores the diversity in the public's perception of ESGwashing and sentiment towards entities' sustainability efforts. This suggests that while some entities are perceived positively, others may be facing challenges in either their sustainability practices or in how these practices are communicated to the public. The relatively low but varied Sentiment Polarity and Subjectivity scores further suggest that sentiments expressed are generally neutral but can sometimes be subjective or strongly polarized, which might affect stakeholder trust and engagement.

4.1. Correlation between scores

The scatter plots (Fig. 2) explore the relationship between sentiment and ESGwashing, and sustainability and ESGwashing. There appears to be a pattern indicating that higher sustainability scores and more positive sentiment scores are associated with lower ESGSI values. This could imply that entities with better sustainability practices and more positive public sentiment are less likely to engage in ESGwashing. In analyzing the scatter plots, we make several observations. The Normalized Sentiment Score plot shows a distribution of points that do not suggest a clear or strong linear relationship with the ESGSI, implying that sentiment is multifaceted and influenced by a complex set of factors beyond the scope of ESGwashing alone.

On the other hand, the Normalized Sustainability Score plot presents a denser concentration of points towards the higher end of the sustainability score, yet still spread across the ESGSI. This suggests that while higher sustainability scores are common, their association with ESGwashing is less predictable, and they do not uniformly correspond to lower ESGSI values.

The absence of a definitive trend in both plots could be indicative of a more nuanced relationship between these factors. Entities with high sustainability scores can still be perceived as engaging in ESGwashing, potentially due to public skepticism or the intricacies of their sustainability reporting. Conversely, entities with lower sustainability scores are not uniformly associated with higher ESGwashing, possibly due to varied expectations and interpretations of what constitutes both sustainability and ESGwashing in different contexts.

These insights highlight the complexity of interpreting sustainability and sentiment data and underscore the need for a more detailed analysis to understand the driving forces behind these relationships. It is also a reminder of the importance of transparent and consistent sustainability reporting to foster a more informed and less skeptical public perception.

4.2. ESGSI across corporate characteristics

4.2.1. ESGSI by geographical region

Table 3 provides an overview of descriptive statistics on sustainability metrics across different regions. It reveals variations in ESGSI, sustainability scores, sentiment scores, polarity, and subjectivity. These metrics are critical for understanding regional differences in the perception and implementation of sustainability practices.

The data illustrates that South America has the lowest sentiment scores and highest negative ESGSI, which might suggest a stronger perception of ESGwashing or less favorable views on sustainability efforts. Conversely, North America shows higher sustainability and sentiment scores, indicating more positive public perception and possibly more effective sustainability initiatives. Europe and Other regions show moderate scores across the metrics but still reflect unique

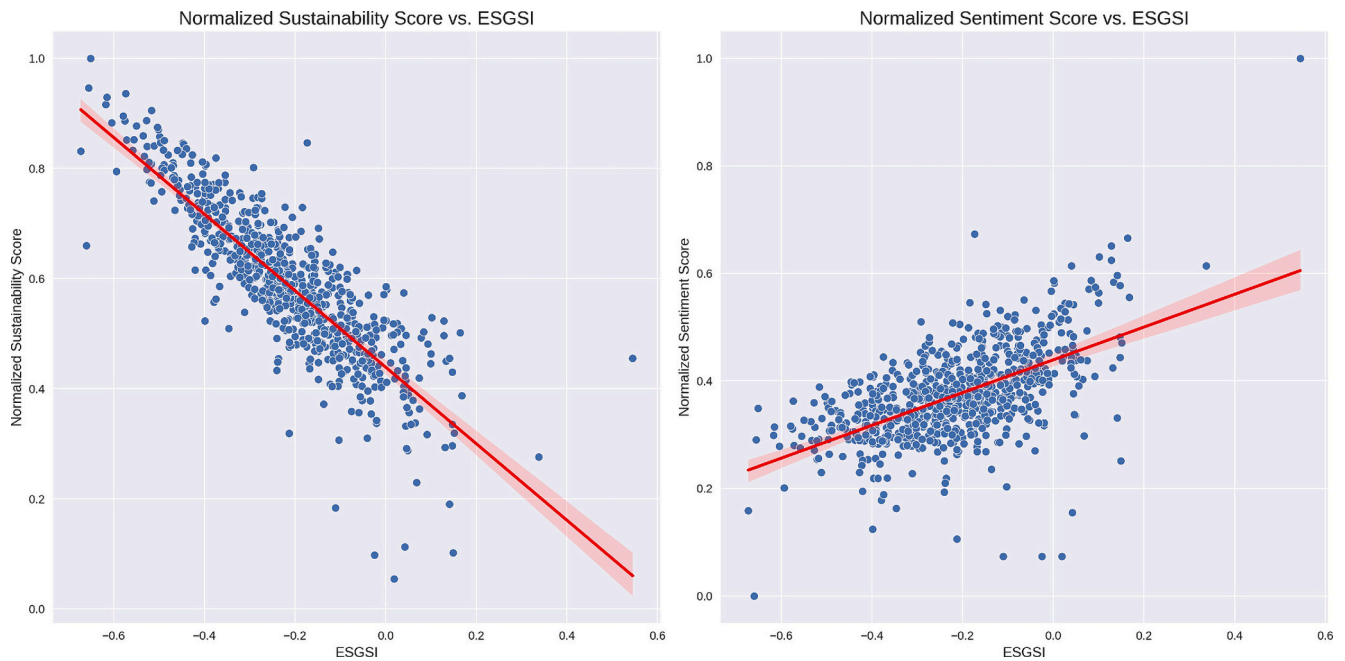


Fig. 2. Scatter Plots of Normalized Sustainability and Sentiment Score vs. ESGSI.

Table 3
Descriptive Statistics by Region.

Region	ESGSI ^a	Normalized Sustainability Score ^b	Normalized Sentiment Score ^c	Sentiment Polarity ^d	Sentiment Subjectivity ^e
Europe	−0.2667	0.6144	0.3477	0.0883	0.3533
North America	−0.1814	0.5731	0.3917	0.1024	0.3498
South America	−0.3393	0.6622	0.3229	0.0803	0.3213
Other	−0.2095	0.5760	0.3665	0.0943	0.3499
Total	−0.2492	0.6064	0.3572	0.0913	0.3436

^a ESGSI: Indicates the level of perceived ESGwashing activities, with lower values suggesting less engagement in such practices.

^b Normalized Sustainability Score: Reflects the sustainability efforts normalized across regions, where higher values denote better performance.

^c Normalized Sentiment Score: Measures overall sentiment towards sustainability efforts, normalized for comparability.

^d Sentiment Polarity: Represents the general tone of sentiment, with higher values indicating more positive sentiment.

^e Sentiment Subjectivity: Indicates the level of subjectivity in the expressed sentiments, with higher values suggesting more bias.

regional challenges and achievements in sustainability. The Total row provides a global overview, suggesting a generally moderate perception of ESGwashing and an overall positive but varied sentiment towards sustainability.

This comprehensive analysis helps in understanding how different

regions tackle sustainability and public sentiment, which can be crucial for global strategies aimed at enhancing corporate sustainability practices.

Table 4
Descriptive Statistics by Industrial Sectors.

Industrial sector	ESGSI ^a	Normalized Sustainability Score ^b	Normalized Sentiment Score ^c	Sentiment Polarity ^d	Sentiment Subjectivity ^e
Communication Services	−0.2272	0.5909	0.3637	0.0934	0.3514
Consumer Discretionary	−0.1802	0.5889	0.4087	0.1079	0.3555
Consumer Staples	−0.2089	0.6138	0.4048	0.1066	0.3589
Energy	−0.3117	0.6595	0.3478	0.0883	0.3589
Financials	−0.2484	0.5944	0.3460	0.0877	0.3398
Health Care	−0.1943	0.5775	0.3832	0.0997	0.3456
Industrials	−0.1893	0.5743	0.3850	0.1003	0.3587
Information Technology	−0.1759	0.5547	0.3788	0.0983	0.3411
Materials	−0.2603	0.6187	0.3584	0.0917	0.3570
Real Estate	−0.2038	0.5672	0.3634	0.0933	0.3410
Utilities	−0.2390	0.5873	0.3483	0.0884	0.3549
Total	−0.2217	0.5934	0.3716	0.0960	0.3512

^a ESGSI: A measure of perceived ESGwashing activities, with lower (more negative) values indicating less ESGwashing.

^b Normalized Sustainability Score: A score normalized between 0 and 1, indicating the extent of sustainable practices within the sector.

^c Normalized Sentiment Score: Represents the normalized sentiment towards the sector's sustainability efforts, scaled between 0 and 1.

^d Sentiment Polarity: Measures the overall tone of sentiment, with higher values indicating more positive sentiments.

^e Sentiment Subjectivity: Reflects how subjective or objective the sentiments are, with higher scores indicating greater subjectivity.

4.2.2. ESGSI by sector

We grouped companies by sector and calculated the mean ESGSI for each sector, as shown in Table 4. The table presents descriptive statistics for various industrial sectors across different metrics, including the ESGSI, Normalized Sustainability Score, Normalized Sentiment Score, Sentiment Polarity, and Sentiment Subjectivity. Each row represents a different industrial sector, such as Consumer Goods, Energy & Utilities, Financial Services, Healthcare, etc. These sectors encompass different industries and sectors of the economy.

The table shows the mean values of the descriptive statistics for each industrial sectors. For example, in the Consumer Goods industrial sectors, the mean ESGSI is -0.3442 , the mean Normalized Sustainability Score is 0.5730 , and so on.

Sectors such as Materials & Chemicals and Energy & Utilities exhibit some of the lowest scores in sustainability but have a more pronounced negative ESGSI. This might suggest that despite lower sustainability scores, these industries are perceived or report themselves as minimally engaging in ESGwashing. This could reflect a significant public or regulatory scrutiny that compels transparent reporting. On the other hand, sectors like Real Estate and Financial Services, which show higher sustainability scores, also exhibit lower ESGwashing indices, indicating potentially better integration of genuine sustainability practices. Brogi & Lagasio, 2018 examine the association between ESG disclosure and company profitability. They find that for financial intermediaries, especially banks, there is a strong relationship between environmental awareness and profitability, which provides significant implications for policy makers and policy takers. Their findings suggest that investments in ESG should be part of a long-term strategy to maintain a positive effect on company performance over time.

The Normalized Sentiment Scores and Sentiment Polarity provide additional layers of insight. Higher sentiment scores in Industrial & Manufacturing and Transportation & Logistics suggest a more favorable public perception, potentially due to effective communication of sustainability efforts or actual impactful practices. Conversely, sectors like Financial Services and Energy & Utilities show lower sentiment scores, which could be due to public skepticism or recent scandals affecting public perception.

Finally, Sentiment Subjectivity scores are relatively stable across sectors, suggesting that the reporting or discourse around these sectors maintains a consistent level of objectivity, which is crucial for ensuring that assessments of sustainability are based on factual reporting rather than opinion.

4.2.3. ESGSI by size

The descriptive statistics by asset group are presented in Table 5.

The table allows for a comparison of the asset groups based on these various metrics. Lower ESGSI values suggest less ESGwashing within that group. Specifically, we observe that the mean ESGSI varies across the asset groups, with lower values in the 1st quartile compared to the others. This suggests that companies in the 1st quartile may exhibit

lower levels of ESGwashing behavior compared to the others. Higher Normalized Sustainability and Sentiment Scores indicate potentially more sustainable practices and positive sentiment, respectively. Notably, the ESGSI remains negative across all quartiles, suggesting a low prevalence of ESGwashing within these groups. This observation could imply either effective regulatory compliance or a genuine commitment to sustainability practices by the organizations in these groups. The Normalized Sustainability Score maintains a range between 0.55 and 0.58 , reflecting a moderate to high level of sustainability, which could be indicative of successful integration of sustainable practices in core business strategies. Sentiment Polarity helps understand the overall sentiment leaning (positive or negative) within the text associated with each group. The Normalized Sentiment Score and Sentiment Polarity indicate a generally positive sentiment towards these asset groups, albeit with a slight decline in the third and fourth quartiles. This may suggest a correlation between higher scrutiny or expectations and slightly less positive public sentiment as sustainability efforts become more mature or visible. Finally, Sentiment Subjectivity reflects the level of opinion or personal judgment in the text. Sentiment Subjectivity scores remain consistently low, suggesting that the discussions or reports analyzed are relatively objective, which is crucial for maintaining credibility in sustainability reporting.

4.2.4. ESGSI by ESG score

Table 6 provides a breakdown of the descriptive statistics for the ESGSI, Normalized Sustainability Score, Normalized Sentiment Score, Sentiment Polarity, and Sentiment Subjectivity, grouped by ESG performance quartiles. This table delineates the statistics within quartiles based on ESG performance. Each quartile represents a subset of the dataset categorized by their respective ESG scores, from the highest (1st Quartile) to the lowest (4th Quartile). The statistics indicate trends and variances in how different ESG groupings correlate with the ESGSI, sustainability scores, and sentiment measures.

From the data, we observe a tendency for the ESGSI to decrease (more negative) as ESG performance decreases (from 1st to 4th Quartile). This suggests that entities with lower ESG ratings may be perceived as more likely to engage in ESGwashing. Additionally, the highest ESG performers (1st Quartile) have the highest Normalized Sustainability Scores and Sentiment Scores, indicative of stronger sustainability practices and more favorable public sentiment.

Conversely, as we move towards the lower quartiles, there is a slight decline in both sustainability and sentiment scores, albeit with a relatively stable Sentiment Polarity across groups. This pattern highlights potential challenges faced by entities with lower ESG scores (collected from Refinitiv), possibly reflecting less effective communication or implementation of sustainability initiatives.

The stability in Sentiment Subjectivity across quartiles suggests a consistent level of objectivity in how sentiments are expressed, irrespective of ESG performance. This uniformity in subjectivity levels provides a neutral basis for comparing sentiment across groups,

Table 5

Descriptive Statistics by Asset Group.

Asset Group	ESGSI	Normalized Sustainability Score	Normalized Sentiment Score	Sentiment Polarity	Sentiment Subjectivity
	(a)	(b)	(c)	(d)	(e)
1st Quartile	-0.1835	0.5688	0.3853	0.1004	0.3566
2nd Quartile	-0.2049	0.5878	0.3829	0.0996	0.3486
3rd Quartile	-0.2329	0.5949	0.3620	0.0929	0.3474
4th Quartile	-0.2498	0.6093	0.3595	0.0921	0.3484
Total	-0.2178	0.5902	0.3724	0.0962	0.3503

Note: This table summarizes the key characteristics (descriptive statistics) for each of the four asset groups analyzed in the study. The statistics reported include: ESGSI: This index likely ranges from negative values (indicating no ESGwashing) to positive values (indicating strong ESGwashing). Normalized Sustainability Score: This score is likely normalized between 0 (low sustainability) and 1 (high sustainability). Normalized Sentiment Score: This score is likely normalized between 0 (low sentiment) and 1 (high sentiment). Sentiment Polarity: This value indicates the overall sentiment (positive, negative, or neutral) of the associated text. It may range from negative to positive, with 0 indicating neutral sentiment. Sentiment Subjectivity: This value reflects the subjectivity of the associated text. It may range from 0 (objective) to 1 (highly subjective).

Table 6
Descriptive Statistics by ESG Group.

ESG Group	ESGSI ^a	Normalized Sustainability Score ^b	Normalized Sentiment Score ^c	Sentiment Polarity ^d	Sentiment Subjectivity ^e
1st Quartile	−0.1793	0.5650	0.3857	0.1005	0.3564
2nd Quartile	−0.2127	0.5898	0.3772	0.0977	0.3475
3rd Quartile	−0.2104	0.5855	0.3751	0.0971	0.3489
4th Quartile	−0.2695	0.6223	0.3529	0.0899	0.3495
Total	−0.2180	0.5907	0.3727	0.0963	0.3506

^a ESGSI: A metric indicating the level of perceived ESGwashing practices. Lower (more negative) values suggest lesser engagement in ESGwashing.

^b Normalized Sustainability Score: A score from 0 to 1, where higher values indicate better sustainability practices.

^c Normalized Sentiment Score: Reflects the general sentiment towards the entity's sustainability efforts, normalized between 0 and 1.

^d Sentiment Polarity: Indicates the overall tone of sentiment; positive values suggest a more positive sentiment.

^e Sentiment Subjectivity: Measures the level of subjectivity in sentiment expressions, with higher values indicating more subjective views.

emphasizing that the perceived differences in sustainability and ESGwashing practices are not due to biases in sentiment expression but likely reflect genuine differences in corporate behavior and public perception.

4.2.5. ANOVA tests

We further employed one-way ANOVA to test for significant differences in ESGSI scores across categories for each characteristic. Post-hoc Tukey's Honest Significant Difference (HSD) tests were conducted to identify specific group differences. Effect sizes were calculated using eta-squared (η^2).

Table 7 presents the ANOVA results for all analyzed corporate characteristics.

Our analysis reveals that the ESGSI is influenced by various corporate characteristics, with geographic location and industry sector showing the strongest effects.

The analysis by Continent shows the strongest effect ($\eta^2 = 0.0746$) of geographic location on ESGSI. European companies tend to have significantly lower ESGSI scores compared to North American companies (mean difference = 0.0853, $p = 0.05$), while South American companies have the highest ESGSI scores. These findings suggest that ESGwashing practices may be influenced by regional regulatory environments and cultural attitudes towards ESG issues. The stricter ESG regulations in Europe might be contributing to lower ESGSI among European companies.

The GICS Sector analysis shows a moderate effect ($\eta^2 = 0.0512$) of industry sector on ESGSI. Post-hoc analysis indicates that the Energy sector has significantly higher ESGSI scores compared to Health Care (mean difference = 0.1174, $p = 0.05$) and Industrials (mean difference = 0.1224, $p = 0.05$) sectors. This suggests that companies in sectors with higher environmental impacts (e.g., Energy) may be more prone to ESGwashing practices, possibly due to greater pressure to appear environmentally responsible.

The analysis shows a smaller effect of company size on ESGSI ($\eta^2 = 0.0249$ for Assets Category). Companies with High asset levels have significantly higher ESGSI scores compared to those with Low asset levels (mean difference = 0.0493, $p = 0.05$), but the effect is not as strong as in our earlier analysis.

The analysis of ESG Score Category on ESGSI ($\eta^2 = 0.0407$) indicates a small-to-moderate relationship between reported ESG performance and ESGSI. Companies in the Very High ESG Score category have significantly lower ESGSI scores compared to those in the High (mean difference = −0.059, $p = 0.05$) and Low (mean difference = −0.0902, p

= 0.05) categories. This suggests that companies with the highest reported ESG scores are less likely to engage in ESGwashing practices, potentially indicating a more genuine commitment to ESG principles.

5. Conclusion

This study introduces the ESGwashing Severity Index (ESGSI), a novel metric that represents a significant advancement in quantifying corporate ESGwashing practices. By juxtaposing the positivity of sentiment with substantive sustainability content in corporate reports, the ESGSI offers a nuanced assessment of the alignment between companies' sustainability claims and their actual practices.

Our analysis reveals several key findings that contribute to the understanding of ESGwashing practices. First, geographic location emerged as the strongest predictor of ESGSI, with European companies demonstrating lower ESGSI scores compared to their North American counterparts. This finding underscores the critical role of regional regulatory environments and cultural attitudes in shaping corporate sustainability practices.

Second, industry sector significantly influences ESGSI scores, with sectors facing higher environmental scrutiny, such as Energy, showing a greater propensity for ESGwashing. This suggests that external pressures may inadvertently contribute to the misalignment between sustainability claims and practices in certain industries.

Third, company size still influences ESGwashing practices, however its effect is not very pronounced. This indicates that ESGwashing is a concern across the corporate spectrum, not limited to larger entities with more resources for sophisticated communication strategies.

Fourth, companies with the highest ESG scores tend to have lower ESGSI values, suggesting a potential link between genuine sustainability commitment and more accurate reporting practices.

These findings have significant implications for various stakeholders. For investors, the ESGSI provides a valuable tool for due diligence, particularly when evaluating companies in high-impact sectors or regions with less stringent ESG regulations. For corporate managers, our results highlight the importance of aligning sustainability communications with tangible practices to build genuine ESG credibility. Policy-makers can leverage the ESGSI to inform the development of more effective ESG reporting standards and regulatory frameworks, potentially using it as a benchmarking tool to identify best practices and areas requiring intervention.

The ESGSI's potential extends beyond its immediate applications. It can serve as a catalyst for enhancing the overall integrity of sustainability reporting, driving companies towards more transparent and authentic ESG practices. By providing a quantifiable measure of the gap between sustainability rhetoric and reality, the ESGSI can contribute to the ongoing evolution of corporate sustainability standards and accountability mechanisms.

However, this study also has limitations that point to promising avenues for future research. The cross-sectional nature of our data limits causal inferences, highlighting the need for longitudinal studies to examine how changes in corporate characteristics and regulatory

Table 7
ANOVA Results for ESGSI by Corporate Characteristics.

Characteristic	F-statistic	p-value	η^2	Interpretation
GICS Sector	3.979.157	2,58E-02	0.0512	Moderate effect
Assets Category	6.165.754	3,86E-01	0.0249	Small effect
ESG Score Category	10.534.781	8,61E-04	0.0407	Small-to-moderate
Continent	20.022.260	1,73E-09	0.0746	Moderate effect

environments affect ESGSI over time. Future research could also delve deeper into the specific ESG practices that are most prone to washing, investigate the mechanisms behind geographic differences in ESGSI scores, and explore potential non-linear relationships between ESG performance and ESGSI.

Furthermore, the development and validation of more sophisticated ESGSI measures could enhance our ability to detect and understand nuanced forms of ESGwashing across different sectors and company sizes. This could involve incorporating additional data sources, refining sentiment analysis techniques, or developing sector-specific ESGSI variants.

In conclusion, the ESGSI represents a transformative approach to assessing and understanding the ESGwashing phenomenon. It holds significant potential to drive change across the corporate landscape, promoting a more transparent and sustainable global economy. As sustainability continues to take center stage in corporate strategy and investor decision-making, tools like the ESGSI will play a crucial role in ensuring that corporate sustainability efforts are genuine, impactful, and aligned with stakeholder expectations. By bridging the gap between sustainability claims and practices, the ESGSI can contribute to building a more trustworthy and effective framework for corporate sustainability, ultimately supporting the transition towards a more sustainable and resilient global economic system.

Data availability

No

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